BASE CHANGE IS A FUNCTOR ON THE FM KERNEL CATEGORY

THE STACKS PROJECT AUTHORS

Editorial note. Closing sentence 'Some details omitted.' covers canonical-isomorphism bookkeeping after the main base-change computation; transcribed as-is.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 3778 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Flat proper base change. perfect-lemma-flat-proper-perfect-direct-image-general (cross-chapter, named). Identifies the pullback of the convolution with the convolution of the pullbacks.

Source reference. Original reference 0B91: focused-seeds/geometry/S9-eqv-base-change-is-functor/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $S' \to S$ be a morphism of schemes. The rule which sends*

(1) *a smooth proper scheme X over S to $X' = S' \times_S X$, and*
(2) *the isomorphism class of an object K of $D_{perf}(\mathcal{O}_{X \times_S Y})$ to the isomorphism class of $L(X' \times_{S'} Y' \to X \times_S Y)^* K$ in $D_{perf}(\mathcal{O}_{X' \times_{S'} Y'})$*

is a functor from the category defined for S to the category defined for S'.

Proof. To see this suppose we have X, Y, Z and $K \in D_{perf}(\mathcal{O}_{X \times_S Y})$ and $M \in D_{perf}(\mathcal{O}_{Y \times_S Z})$. Denote $K' \in D_{perf}(\mathcal{O}_{X' \times_{S'} Y'})$ and $M' \in D_{perf}(\mathcal{O}_{Y' \times_{S'} Z'})$ their pullbacks as in the statement of the lemma. The diagram

$$\begin{array}{ccc} X' \times_{S'} Y' \times_{S'} Z' & \longrightarrow & X \times_S Y \times_S Z \\ \downarrow{\scriptstyle \mathrm{pr}'_{13}} & & \downarrow{\scriptstyle \mathrm{pr}_{13}} \\ X' \times_{S'} Z' & \longrightarrow & X \times_S Z \end{array}$$

is cartesian and pr_{13} is proper and smooth. By Derived Categories of Schemes, Lemma 0B91 we see that the derived pullback by the lower horizontal arrow of the composition

$$R\mathrm{pr}_{13,*}(L\mathrm{pr}_{12}^* K \otimes^{\mathbf{L}}_{\mathcal{O}_{X \times_S Y \times_S Z}} L\mathrm{pr}_{23}^* M)$$

indeed is (canonically) isomorphic to

$$R\mathrm{pr}'_{13,*}(L(\mathrm{pr}'_{12})^* K' \otimes^{\mathbf{L}}_{\mathcal{O}_{X' \times_{S'} Y' \times_{S'} Z'}} L(\mathrm{pr}'_{23})^* M')$$

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as desired. Some details omitted. □